Lecture 3 - Fermions on a lattice

Simran Singh

May 5, 2026

Abstract

In this lecture we will see the challenges one faces when placing fermionic degrees of freedom on the lattice. In order to understand the problem, we will first have a brief look at continuum Minkowski fermions, and see what changes when we go to Euclidean spacetime. With this we will also recap on Grassmann algebra and the Dirac equation. We then begin by naively discretizing the Dirac equation in Euclidean spacetime and see what differs from the scalar theory - as seen from the propagator. This will lead us to the doubling problem and the famous Nielsen-Ninomiya theory - at least a simplified version of it. We will then briefly see how one must go beyond the naive discretization and only introduce the Wilson and Staggered fermions - you will hear more about this in a later lecture.

1 Overview

In order to get a complete description of nature, fermionic fields theories need to be considered. Fermions - unlike gauge or scalar fields - have no classical interpretation. Generally the first time one hears about fermions is in an introductory QFT lecture (or in quantum statistical mechanics when discussing the spin-statistics theorem). Usually, this is introduced as an attempt (by Paul Dirac) to create a version of the Schrödinger equation that also satisfies Lorentz invariance (as any quantum field theory should). This then leads to the discovery of a linear equation in space and time with some additional *spin* structure that naturally describes spin-1/2 particles. Further, if one tries to quantize the Dirac equation, one finds that the usual commutation relations which were enough to describe scalars and gauge fields - now lead to inconsistencies. One problem being that if one insists on using commutators, one is lead to an unbounded Hamiltonian. In order to re-concile all we know about spin half particles (Pauli exclusion principle) and not have unbounded Hamiltonians, we are forced to use *anti-commutators* to describe fermions.

I highly recommend reading Chapters 4 & 5 in David Tong's lecture notes[1] on QFT

As we will discuss in today's lecture, it is quite hard to place fermions on a discrete spacetime lattice unless we are willing to break a few symmetries (Chiral) or meaningful physical requirements like locality. If we naively try to discretize free fermions like we did for the free scalar fields (comparing matter fields to matter fields), we run into an infamous problem called the "doubling problem". Essentially this means that if we try to discretize a theory of a single free fermion, and then try to remove this discretization by taking the continuum limit - we end up with a *different* theory - one with twice the number of fermions per dimension than the original theory. For these reasons lattice fermions are still an active area of research.

In this lecture we will briefly motivate two fermion discretizations beyond the naive version: Staggered and Wilson fermions. Since this is a very delicate topic - you will hear it once again in the 9th lecture from the Hamiltonian perspective.

Disclaimer: The purpose of this lecture is *not* to introduce fermions rigorously as this is done over many weeks in a proper QFT course! However since our goal is to *put fermions on the lattice* - we will need to introduce the *bare minimum* continuum knowledge needed for this.

2 Continuum Fermions & Grassmann Algebra: A Recap

In this section we briefly remind ourselves about some properties of fermions in the continuum and in Minkowski spacetime.

The Electron Field and Its Indices

To begin, let us consider the simplest fermionic field theory – *Quantum Electrodynamics* (QED), which describes electrons interacting with photons – but without the gauge fields. The key difference from the scalar field we studied previously is that the fermionic field $\psi(x)$ is *not* a single number at each spacetime point - it is a multi-component object called the *Dirac spinor field*. For an electron, the field carries a single type of index,

For QCD (quarks) one adds a colour index $a \in \{r, g, b\}$ and a flavour index $f \in \{u, d, s, c, b, t\}$, giving the full field $\psi^a_{\alpha,f}(x)$.

$$\psi_\alpha(x)\,, \qquad \alpha \in \{1, 2, 3, 4\}\,, \tag{1}$$

where the indices have the following meaning,

$$\begin{aligned} x \quad &: \quad \text{spacetime point, as usual}\,, \\ \alpha \in \{1,2\} \quad &: \quad \text{electron components (spin-up and spin-down)}\,, \\ \alpha \in \{3,4\} \quad &: \quad \text{positron components (spin-up and spin-down)}\,. \end{aligned} \tag{2}$$

Note that for the electron there is no colour index - electrons do not carry colour charge and so are blind to the strong force. There is also no flavour index since there is only one type of electron. The field $\psi_\alpha(x)$ takes values in $\mathbb{C}^4$ - it is a 4-component complex-valued spinor at each spacetime point.

One also needs the Dirac conjugate field, defined as,

The conjugate field is defined in order to get a Lorentz scalar - without the γ_0, we would not get a scalar (see Chapter 4 in David Tong's lecture notes[1] on QFT)

$$\bar{\psi}_\alpha(x) \equiv \sum_{\beta=1}^{4} \psi^\dagger_\beta(x)\,(\gamma^0)_{\beta\alpha}\,, \tag{3}$$

where $\psi^\dagger_\beta$ is the complex conjugate transpose of ψ_β - it is a row vector with index β - and $(\gamma^0)_{\beta\alpha}$ is the (β, α) matrix element of γ^0. The sum over β is just the usual matrix-vector multiplication $\psi^\dagger\gamma^0$, and in the standard Einstein summation convention the $\sum_\beta$ is left implicit. where γ^0 is the time-like Dirac matrix. It is important to note that ψ and $\bar{\psi}$ are treated as *independent* Grassmann-valued fields in the path integral.

The spinor index α is contracted with the Dirac matrices $(\gamma^\mu)_{\alpha\beta}$, which are 4×4 matrices satisfying the *Clifford algebra*,

$$\{\gamma^\mu, \gamma^\nu\} \equiv \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\,\eta^{\mu\nu}\,, \tag{4}$$

where $\eta^{\mu\nu} = \mathrm{diag}(+1,-1,-1,-1)$ is the metric of Minkowski spacetime. One last ingredient that we need is the γ^5 matrix and its properties:

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$$
$$\{\gamma^5,\gamma^\mu\} = 0\,.$$

Dirac Equation

The Dirac equation for a free massive fermion is given by,

$$(i\gamma^\mu\partial_\mu - M)\,\psi(x) = 0\,, \tag{5}$$

while the Dirac action is given by,

$$S[\psi,\bar\psi] = \int d^4x\,\bar\psi(x)\,(i\gamma^\mu\partial_\mu - M)\,\psi(x)\,. \tag{6}$$

Grassmann Algebra

The most important aspect of fermion fields is that they satisfy the Grassmann algebra - necessary to ensure the Fermi statistics . This means that if we consider a product of two fermion fields, they satisfy,

Hermann Günther Grassmann (1809–1877) was a German mathematician and linguist who introduced what we now call exterior algebra in his 1844 work *Die lineale Ausdehnungslehre*. His work was largely ignored during his lifetime and only recognised after his death. His work was later found to be exactly the right mathematical structure to describe fermions in quantum field theory.

$$\psi(x)_\alpha\psi(x')_\beta = -\psi(x')_\beta\psi(x)_\alpha\,. \tag{7}$$

This is called the *anti-commutativity* property of Grassmann numbers. In general, a set of Grassmann numbers $\{\theta_i\}$ with $i \in \{1,2,\ldots,N\}$ are defined by the anticommutation relations,

$$\{\theta_i,\theta_j\} = \theta_i\theta_j + \theta_j\theta_i = 0 \quad \text{for all } i,j\,. \tag{8}$$

An element of the Grassmann algebra is a polynomial in the generators,

$$f(\theta) = f + \sum_i f_i\theta_i + \sum_{ij} f_{ij}\theta_i\theta_j + \sum_{ijk} f_{ijk}\theta_i\theta_j\theta_k + \cdots\,, \tag{9}$$

where the coefficients $f_{ij\ldots l}$ are ordinary complex (or sometimes real) numbers, which are antisymmetric in $i,j,\ldots,l$. Setting $i = j$ in the anticommutation relation immediately gives us a very important consequence - the square of any Grassmann number is zero,

$$\theta_i^2 = 0\,. \tag{10}$$

This means the polynomial above terminates at finite order and, in particular, the Taylor expansion of any function $f(\theta)$ of a single Grassmann variable truncates after the first order,

$$f(\theta) = f_0 + f_1\theta\,, \tag{11}$$

where f_0 and f_1 are ordinary (commuting) numbers. One also defines formal differentiation and integration procedures. The differentiation rules are,

$$\frac{\partial}{\partial\theta_i}\theta_i = 1\,, \qquad \frac{\partial}{\partial\theta_i}\theta_i\theta_j = \theta_j\,, \qquad \frac{\partial}{\partial\theta_i}\theta_j\theta_i = -\theta_j\,, \tag{12}$$

and integration is defined by the *Berezin integral*,

$$\int d\theta \, 1 = 0 \,, \qquad \int d\theta \, \theta = 1 \,. \tag{13}$$

This will be relevant in the exercises when you compute the fermion path integral!

Note that Grassmann numbers are not real or complex numbers in the usual sense - they are purely formal algebraic objects. In practice, one never needs to assign a numerical value to θ - what matters is the algebra they satisfy. Finally, a Grassmann number anti-commutes with all other Grassmann numbers but commutes with ordinary numbers,

$$\theta_i \, c = c \, \theta_i \,, \qquad \theta_i \, \theta_j = -\theta_j \, \theta_i \,, \tag{14}$$

where $c \in \mathbb{C}$ is an ordinary number. These rules are everything one needs to define the fermion path integral, which we will come to shortly.

3 Euclidean Continuum Fermions

In this section we will motivate the conventions for the Euclidean Dirac matrices. In order to do this recall that we want to go from the oscillatory path integral in $S[\psi, \bar{\psi}]$ to a well defined Euclidean version which has a probabilistic interpretation (i.e. a positive-definite Boltzmann weight[1]),

$$\int \mathcal{D}\bar{\psi}\mathcal{D}\psi \, e^{iS[\psi,\bar{\psi}]} \xrightarrow{t \to -i\tau} \int \mathcal{D}\bar{\psi}\mathcal{D}\psi \, e^{-S_E[\psi,\bar{\psi}]} \,. \tag{15}$$

So we need two things: (i) a real action, (ii) and overall negative sign in front of a positive action. In order to do this we follow the prescription,

$$\gamma_4^E \quad \to \quad \gamma^0 \tag{16}$$
$$\gamma_i^E \quad \to \quad -i\gamma^i \,. \tag{17}$$

This gives us the Euclidean version of the Clifford algebra in Eq. (4),

$$\{\gamma_\mu^E, \gamma_\nu^E\} = 2 \, \delta_{\mu\nu} \,. \tag{18}$$

A key consequence of this prescription is that all Euclidean gamma matrices are Hermitian, $(\gamma_\mu^E)^\dagger = \gamma_\mu^E$, in contrast to the Minkowski case where $(\gamma^i)^\dagger = -\gamma^i$. With this prescription, the Euclidean Dirac action becomes,

$$S^E[\psi, \bar{\psi}] = \bar{\psi}(x) \left(\gamma_\mu^E \partial_\mu + M \right) \psi(x). \tag{19}$$

Note that although $(\gamma_\mu^E)^\dagger = \gamma_\mu^E$, the Dirac operator $\not{\partial}_E = \gamma_\mu^E \partial_\mu$ is anti-Hermitian, $(\not{\partial}_E)^\dagger = -\not{\partial}_E$, since $\partial_\mu^\dagger = -\partial_\mu$. Its eigenvalues are therefore purely imaginary, $\pm i\lambda_n$ with $\lambda_n \in \mathbb{R}$. The eigenvalues of the operator including the mass terms are then of the form $i\lambda + M$ (and the complex conjuagtes as we argue below).

Contrast this with the case of the free scalar field, for which we had a well-defined configuration $\phi(x)$ with the probability weight $e^{-S^E[\phi(x)]}$!

As one can guess, the positive-definiteness of the Boltzmann weight cannot be argued pointwise for fermions, since ψ and $\bar{\psi}$ are Grassmann variables and e^{-S_E} is

[1] For fermions, ψ and $\bar{\psi}$ are Grassmann variables, so $e^{-S_E[\psi,\bar{\psi}]}$ is not a real number configuration-by-configuration and positivity must be argued differently, as we discuss below.

not a positve number for each field configuration. Instead, one integrates out the fermions exactly to obtain a functional determinant,

$$\int \mathcal{D}\bar{\psi}\mathcal{D}\psi \; e^{-\bar{\psi}(\not{\partial}_E+M)\psi} = \det(\not{\partial}_E + M). \tag{20}$$

In order to understand the reality and positivity of this determinant, we need to look at the spectrum of the matrix $D \equiv \not{\partial}_E + M$. As we have seen the Euclidean Dirac operator does not seem to have any hermitian properties on its own - but it has something called the γ^5-*hermiticity*. This is the property,

$$\gamma^5 D = D^\dagger \gamma^5 \,. \tag{21}$$

This property ensures that if the Dirac operator has an eigenvector with a complex eigenvalue with $D|u\rangle = \kappa|u\rangle$, then it also has another eigenvector $\gamma^5|u\rangle$ with the eignevalue κ^* - so the eigenvalues of $(\not{\partial}_E + M)$ pair up as $(M + i\lambda_n)(M - i\lambda_n)$, giving

$$\det(\not{\partial}_E + M) = \prod_n \left(M^2 + \lambda_n^2\right) > 0\,! \tag{22}$$

Thus the Boltzmann weight is positive-definite and the path integral is well-defined. The γ^5-hermiticity was essential to counter the anti-Hermiticity of $\not{\partial}_E$: it forces the eigenvalue spectrum to consist of complex conjugate pairs, guaranteeing positivity.

4 Naive Lattice Fermions: Doubling Problem

After having discussed the Euclidean version of the Dirac equation, we are now ready to introduce a spacetime lattice and formulate the lattice free fermion theory. We will see that the treatment follows very similarly to the case of the scalar field (both being a matter fields) - and the unusual problems only occuring when we think about the continuum limit. We begin by discussing the dimensionless fields we will put on the lattice:

$$\begin{aligned}
\hat{M} &= aM && (23)\\
\hat{\psi}_\alpha(n) &= a^{3/2}\psi(x) && (24)\\
\hat{\bar{\psi}}_\alpha(n) &= a^{3/2}\bar{\psi}(x) && (25)\\
\hat{\partial}_\mu\hat{\psi}_\alpha(n) &= a^{5/2}\partial_\mu\psi_\alpha(x)\,. && (26)
\end{aligned}$$

As we have seen in the lecture on lattice scalar fields, there are many ways to discretise the derivative on the lattice using the forward, backward and symmetric definitions,

$$\hat{\partial}_\mu\hat{\psi}_\alpha(n) = \begin{cases} \Delta^f_\mu\hat{\psi}_\alpha(n) \equiv \hat{\psi}(n+\hat{\mu}) - \hat{\psi}(n)\,, \\ \Delta^b_\mu\hat{\psi}_\alpha(n) \equiv \hat{\psi}(n) - \hat{\psi}(n-\hat{\mu})\,, \\ \Delta^s_\mu\hat{\psi}_\alpha(n) \equiv \dfrac{\hat{\psi}(n+\hat{\mu}) - \hat{\psi}(n-\hat{\mu})}{2}\,. \end{cases} \tag{27}$$

Notice that among the different definitions, only the symmetric definition has the anti-Hermitian properties that we need for our Euclidean Dirac operator ∂^E_μ.

Once we have identified the lattice derivative that we have to work with, we can write down the dimensionless (naive) free lattice fermion action as,

The following treatment leading upto the doubling problem follows the discussion in Chapter 4 of ref.[2] very closely.

$$S^E[\hat{\psi},\bar{\hat{\psi}}] = \sum_{n,\alpha,\mu>0} \bar{\hat{\psi}}_\alpha(n) \left(\sum_\beta (\gamma^E_\mu)_{\alpha\beta} \left[\frac{\hat{\psi}_\beta(n+\hat{\mu}) - \hat{\psi}_\beta(n-\hat{\mu})}{2} \right] + \hat{M}\hat{\psi}_\alpha(n) \right) , \quad (28)$$

where the spinor indices $\alpha\,,\beta$ have been made explicit. Following the same logic as the scalar field, we can write this action in another form more amenable to computing the partition function,

$$S^E[\hat{\psi},\bar{\hat{\psi}}] = \sum_{\substack{n,m\\ \alpha,\beta}} \bar{\hat{\psi}}_\alpha(n) K_{\alpha\beta}(n,m) \hat{\psi}_\beta(m)\,, \quad (29)$$

where the matrix elements of $K_{\alpha\beta}(n,m)$ are given by,

$$K_{\alpha\beta}(n,m) = \sum_\mu \frac{1}{2} (\gamma^E_\mu)_{\alpha\beta} \left[\delta_{m,n+\hat{\mu}} - \delta_{m,n-\hat{\mu}}\right] + \hat{M}\delta_{m,n}\,. \quad (30)$$

Using this, the quantity that we want to compute is the two-point Euclidean Green's function $\langle \hat{\psi}_\alpha(n) \bar{\hat{\psi}}_\beta(m) \rangle$ – which gives us information on the particle spectrum of our theory. As seen from the case of the scalar field theory, this two-point function can be obtained from a generating functional. The difference now will be (i) spinor indices, and (ii) Grassmann valued sources terms for the fields,

$$Z[\eta,\bar{\eta}] = \int \mathcal{D}\bar{\hat{\psi}} \mathcal{D}\hat{\psi}\; e^{-S^E[\hat{\psi},\bar{\hat{\psi}}] + \sum_{n,\alpha} \left[\bar{\eta}_\alpha(n)\hat{\psi}_\alpha(n) + \bar{\hat{\psi}}_\alpha(n)\eta_\alpha(n) \right]}\,. \quad (31)$$

Carrying out the Grassmann integral, this reduces to,

This is an exercise!

$$Z[\eta,\bar{\eta}] = \det K \exp\left(\sum_{\substack{k,l\\ \lambda,\gamma}} \bar{\eta}_\lambda(k) K^{-1}_{\lambda\gamma}(k,l) \eta_\gamma(l) \right)\,. \quad (32)$$

Finally one can take the derivatives of Eq.(32) with respect to $\eta_\beta(m)$ and $\bar{\eta}_\alpha(n)$, *while carefully following the rules of Grassmann numbers* defined in Eq.(12) to get the two-point function,

$$\langle \hat{\psi}_\alpha(n) \bar{\hat{\psi}}_\beta(m) \rangle = K^{-1}_{\alpha\beta}(n,m)\,. \quad (33)$$

In order to compute the inverse matrix elements in Eq. (33), we will use (i) the fact,

$$\sum_{\lambda,l} K_{\alpha\lambda}(n,l) K^{-1}_{\lambda,\beta}(l,m) = \delta_{m,n}\delta_{\alpha\beta}\,, \quad (34)$$

and using the momentum representation ,

There is an exercise to show the form of the lattice momentum!

$$\langle \hat{\psi}_\alpha(n) \bar{\hat{\psi}}_\beta(m) \rangle = \int_{-\pi}^{\pi} \frac{d^4\hat{p}}{(2\pi)^4} \frac{[-i\gamma_\mu \tilde{\hat{p}} + M]_{\alpha\beta}}{\tilde{\hat{p}}_\mu \tilde{\hat{p}}_\mu + \hat{M}^2}\, e^{i\hat{p}(n-m)}\,. \quad (35)$$

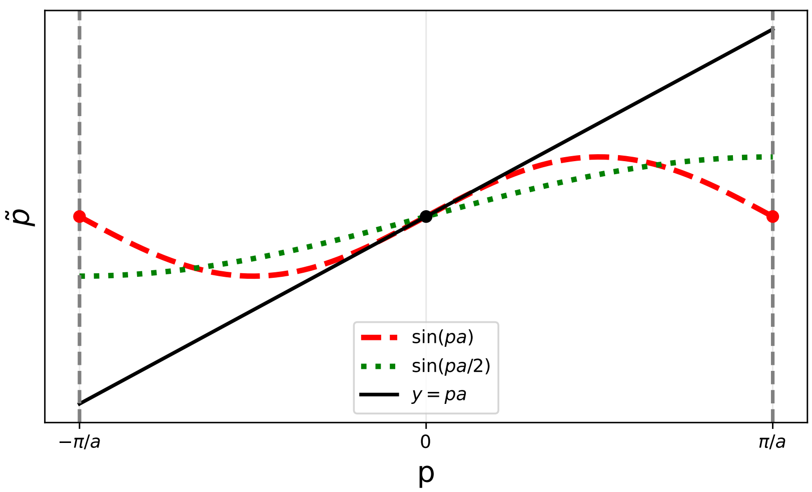

Figure 1: This is an illustration to show the effect of the missing factor of one-half from the fermion lattice momenta in Eq. (36) as opposed to case of the scalar field in Eq. (44) from lecture 1. The The fermionic lattice momentum has zeros at the edges of the Brillouin zone!

In the above, the momentum $\tilde{\tilde{p}}$ – inserting back the factors of lattice spacing – is defined as,

$$\tilde{p}_\mu \equiv \frac{1}{a}\sin(p_\mu a)\,. \tag{36}$$

Finally by re-inserting the appropriate factors of the lattice spacing spacing one gets,

You are encouraged to check the factors!

$$G_{\alpha\beta}(\frac{x}{a},\frac{y}{a},aM) = a^3\int_{-\pi/a}^{\pi/a}\frac{d^4p}{(2\pi)^4}\,\frac{[-i\tilde{\not{p}}+M]_{\alpha\beta}}{\sum_\mu \tilde{p}_\mu\tilde{p}_\mu}\,e^{ip(x-y)} \tag{37}$$

using which one can recover the continuum the two-point function as we did in the scalar field theory,

$$\langle\psi_\alpha(x)\bar{\psi}_\beta(m)\rangle = \lim_{a\to 0}\frac{1}{a^3}G_{\alpha\beta}\left(\frac{x}{a},\frac{y}{a},aM\right)\,. \tag{38}$$

But is this the correct continuum limit? The answer unfortunately is **no** – as opposed to the scalar theory. To see this, begin by comparing Eq. (36) against Eq. (44) from lecture 1, you should notice a big difference – the factor of $\frac{1}{2}$! The problem can be seen in following illustration (Fig. 1), Recalling our arguments for why the continuum limit of the two-point function in case of the scalar field matched the analytical results, that we could replace the lattice momentum $\tilde{p}$ with the continuum version p because only the modes that are small with respect to the inverse lattice spacing contribute to the integral – this doesn't hold true anymore for the fermions due to the missing one-half factor! The modes at the corners of the Brillouin zones contribute equally to the integral in the continuum limit! On a d-dimensional lattice, there are as many continuum-like contributions to the integral

5 An invitation to Non-Naive discretizations: Staggered and Wilson Fermions

6 A No-Go Theorem: Simplified Nielsen-Ninomiya

References

[1] David Tong. Quantum field theory. `https://www.damtp.cam.ac.uk/user/tong/qft.html`, 2006. Lecture notes, University of Cambridge.

[2] Heinz J. Rothe. *Lattice Gauge Theories: An Introduction*, volume 82 of *World Scientific Lecture Notes in Physics*. World Scientific, 4th edition, 2012.